BorNEO HackWknd

TrueCost

Car Affordability Reality Check

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1.0 Introduction

This report introduces **TrueCost**, a tool that helps Malaysians understand the real cost of owning a car. It highlights the problem of hidden expenses, aligns with key SDGs, and shows how the system calculates affordability, long-term financial impact, and offers practical advice.

1.1 Background of Case Study

Malaysia has one of the highest vehicle ownership rates in Asia, with over 33 million registered vehicles for a population of about 33 million people. Due to limited public transportation in many areas, personal vehicles are often considered essential for daily commuting and employment.

However, many consumers evaluate car purchases based mainly on the monthly loan instalment, without considering the full financial impact of ownership. Costs such as fuel, insurance, road tax, maintenance, and depreciation significantly increase the actual monthly expense, often making vehicles far more costly than initially perceived.

As a result, many households underestimate the true cost of vehicle ownership, which can lead to financial strain and poor long-term financial planning.

1.2 Chosen Sustainable Development Goals (SDG)

SDG	Relevance to the Project
SDG 1 – No Poverty	Help prevent households from entering debt due to unaffordable vehicle commitments.
SDG 8 – Decent Work and Economic Growth	Encourages responsible financial behaviour and informed decisions.
SDG 10 – Reduced Inequalities	Provide free access to financial analysis tools that were previously limited to professional financial advisors

1.3 Problem Statement

Buying a car is a big financial decision, but a lot of Malaysians don't have the right tools to figure out if they can really afford it. Most of the time, car dealers and those online loan calculators just show you the monthly payment. They leave out all the other stuff—fuel, insurance, maintenance, and even how much your car loses value over time.

On top of that, there isn't one place where you can get a full picture. There's no single platform that pulls together your personal financial info, calculates the true cost of owning a car, and shows you how it'll affect your finances in the long run. Because of this, plenty of people end up underestimating how much owning a car actually costs, and that can leave them more financially vulnerable and hurt their ability to build wealth over time.

1.4 Project Objectives

The objectives of the TrueCost system are:

1. Calculate the real monthly cost of owning a vehicle by adding together the loan payment, fuel, insurance, road tax, maintenance, and depreciation.
2. Offer a personalized affordability assessment based on the user's financial situation.
3. Show the long-term financial effects of vehicle ownership through an analysis of wealth opportunity costs.
4. Provide financial advice to help users make responsible vehicle choices.
5. Offer an easy-to-use browser tool that lets users check car affordability without needing to register.

2.0 Review of Existing Solutions

Several tools currently exist in the Malaysian market that address partial aspects of car financial planning. The following review evaluates these platforms and identifies the specific gaps that TrueCost is designed to fill.

2.1 Analysis of Existing Tools and Platforms

Table 2.1 : Comparison of Existing Solutions

Platform / Tools	**Primary Function**	**Key Limitations**
Bank & Finance Co. Calculators (E.g. Maybank, CIMB, RHB)	Computes monthly loan instalments based on principal, interest rate and tenure.	Focuses solely on instalments; omits fuel, insurance, maintenance, depreciation and total owner costs. No affordability assessment.
Financial Aggregators (iMoney.my, RinggitPlus.com)	Aggregates and compares car loan packages from various lenders.	Limited to loan comparison; lacks ownership cost breakdowns and personalized financial profiling or AI advice.
Marketplaces (Carlist.my, Mudah.my)	Provides listings for new, reconditioned, and used cars in Malaysia.	Purely transactional; provides no financial analysis, affordability scoring, or running cost data.
Automotive Editorial (Paul Tan's Automotive News)	Offers editorial content on prices, specifications, and industry news.	Informational only; lacks interactive financial tools or personalization.
Generic Budgeting Apps (Mint, Money Lover)	General expense tracking and personal budgeting.	Unable to specify car purchase scenarios or compute specific ownership costs.

2.2 Identified Gaps in the Market

The review of the current landscape reveals several consistent deficiencies across all existing solutions in the Malaysian market. Primarily, there is a significant issue with fragmented cost calculation, as no platform provides a complete true cost ownership that incorporates all five critical components; loan, fuel, insurance/road tax, maintenance, and depreciation in a single view. Furthermore, there is a distinct lack of personalization; existing tools fail to offer financial profile assessments that factor in vital individual data such as debt-to-income (DTI) ratios, job stability, number of dependents and emergency fund adequacy.

Another critical void is the absence of opportunity cost analysis, with no current tool demonstrating the long-term wealth sacrifice or lost investment returns incurred by choosing a more expensive vehicle. Additionally, AI integration is non-existent, meaning tailored financial advice based on a user's specific profile and vehicle choice is entirely unavailable. Ultimately, a major marketplace vs analysis gap persists because existing platforms operates strictly as either pure calculators or pure marketplaces, failing to bridge the gap between browsing for a car and understanding its true financial impact.

2.3 TrueCost's Competitive Differentiation

TrueCost distinguishes itself as the only Malaysian platform that integrates a real 919-vehicle dataset with personalized financial profiling, full ownership cost modeling and wealth gap visualisation. By offering AI-powered advisory in a free, browser based application, it provides a unique value proposition that addresses the information inequality currently facing Malaysian car buyers.

To achieve this , the platform utilizes a specialized technology stack including the Groq API for real-time financial coaching and vehicle recommendations. Unlike traditional calculators, TrueCost bridges the gap between searching for a car and understanding its long-term impact by visualizing the wealth opportunity cost. This feature quantifies the "Track B" scenario, showing users exactly how much wealth they could accumulate over 5 or 10 years by choosing a more affordable vehicle and investing the savings at a 7% annual return. This comprehensive approach transforms the car-buying process from a simple monthly payment calculation into a long-term strategic financial decision.

3.0 Methodology

This section outlines TrueCost's deterministic architecture, detailing the zero-latency frontend, empirical financial models, and the specific data pre-filtering constraints applied to the Groq LLM. These methodologies were explicitly engineered to guarantee mathematical accuracy, eliminate AI hallucinations, and ensure reliable financial advisory.

3.1 Technology Stack

Component	Technology/Details
Frontend	Pure HTML5, CSS3, and Vanilla JavaScript. No frameworks. No build tools. Single-file deployment.
Typography	Google Fonts. "Playfair Display" for headings. "IBM Plex Mono" for data. "IBM Plex Sans" for body text.
AI Inference (In-App)	Groq API. Model chosen is "llama-3.3-70b-versatile" for car recommendations and financial advisory.
Vehicle Dataset	919 different Malaysian cars embedded as a static JSON array. (Note: Implemented as a client-side constraint for the prototype phase to allow for zero-latency filtering and to avoid database hosting dependencies during demonstration purposes.)
Charts & Visualization	Custom SVG line and bar charts rendered programmatically using JavaScript.
Hosting Compatibility	Stateless single-file HTML deployable on any static hosting (Netlify, Vercel, GitHub Pages)

3.2 Application Architecture

TrueCost has been architected as a single-page application with six distinct screens that can be navigated in sequence via a JavaScript-driven step navigation system. There are no server-side elements required apart from the Groq API proxy endpoint.

Step / Screen	Function
Step 1 — Landing Page	Describe the problem: Show the hidden costs through live examples, ticker stats, and cost breakdown tables.
Step 2 — Financial Profile	Gather: Salary, existing loans, expenses, savings, emergency fund, age, dependents, job stability. Compute: Safe budget, maximum car price for 3 different loan tenures, DTI ratio, risk level (SAFE /

	CAUTION / DANGEROUS).
Step 3 — Car Selection	Display the 919 different car dataset in a filterable grid. AI Recommendation Engine will analyze the top results to find the top 3 car matches for the user profile.
Step 4 — True Cost Breakdown	Compute and display all five monthly costs, 5-year total cost of ownership, and income allocation chart.
Step 5 — Wealth Gap Analysis	Two sections: (a) Track A vs Track B SVG line chart displaying the 5-year and 10-year accumulated wealth. (b) 3-Scenario chart (Buy New / Buy Used / Invest the Difference).
Step 6 — AI Financial Advisor	Entire AI analysis section. Interactive chat with 8-message context window. The entire financial profile injected into system prompt.

3.3 Key Calculation Formulas

The following financial models underpin TrueCost's calculations:

Safe Car Budget:

```
const loanTermMonths = (state.loanTermYears||7) * 12;
const disposable = state.salary - state.loans - state.expenses;
const stabilityFactor = state.stability==='high'?1:state.stability==='medium'?0.85:0.7;
const depFactor = Math.max(0.7, 1 - state.dependents * 0.05);
const maxCarPct = 0.20 * stabilityFactor * depFactor;
const budget = state.salary * maxCarPct;
const emergencyTarget = state.expenses * 6;
const emergencyOk = state.emergency >= emergencyTarget;
const freeSavings = Math.max(0, state.savings - state.emergency);
const downPayment = freeSavings * 0.3;
const loanTermYrs = state.loanTermYears||7;
const flatRate = 0.035;
const loanPrincipal = budget * loanTermMonths / (1 + flatRate * loanTermYrs);
const maxPrice = loanPrincipal + downPayment;
const remaining = disposable - budget;
const dti = ((state.loans + budget) / state.salary) * 100;
state.budget = budget; state.dti = dti; state.maxPrice = maxPrice; state.downPayment = downPayment;
```

Figure 3.3.1 The "Demographic Throttling" Formula (Safe Car Budget)

```
const downPayment = car.price*0.1;
const loanAmt = car.price - downPayment;
const rate = isNew ? 0.035 : 0.045;
const months = (state.loanTermYears||7)*12;
const monthly_loan = loanPayment(loanAmt,rate,months);
const fuel = engineCC<=1000?380:engineCC<=1300?480:engineCC<=1500?580:engineCC<=2000?700:900;
const insurance = (car.price*0.024)/12;
const roadtax = engineCC<=1000?20:engineCC<=1300?30:engineCC<=1500?45:engineCC<=2000?70:100;
const maint = isNew?150:220;
const deprec = isNew?(car.price*0.15)/12:(car.price*0.10)/12;
const total = monthly_loan+fuel+insurance+roadtax+maint+deprec;
const total5yr = total*60;
```

Figure 3.3.2 The Amortized Loan Installment Formula

True Monthly Ownership Cost:

Cost Component	Calculation Method
Loan Installment	Flat rate of 3.5% per annum. Principal = Car Price – Down Payment (30% of free savings, capped at 30% of car price). Monthly = (Principal + Total Interest) / Months
Fuel Cost	Engine CC-based consumption rate x Petrol price (RM2.05/L) x 1,500 km/month
Insurance + Road Tax	Insurance = 2.4% of car price / 12 months. Road tax is scaled by engine CC.
Maintenance	RM150–200/month (new), RM250–350/month (used) — scaled by car age
Depreciation	15% of car price per year (new vehicles); 10% per year (used/recond) — divided by 12 for a monthly figure

Wealth Gap (Track B Investment Returns)

```
function compoundFV(monthly, months, annualRate) {
  if (monthly <= 0) return 0;
  const r = annualRate / 12;
  return monthly * ((Math.pow(1+r, months)-1)/r);
}
```

Figure 3.3.3 The Wealth Gap Formula (Opportunity Cost / Track B)

3.4 AI Integration Architecture

TrueCost integrates two distinct AI-powered features, both using the Groq API (llama-3.3-70b-versatile model):

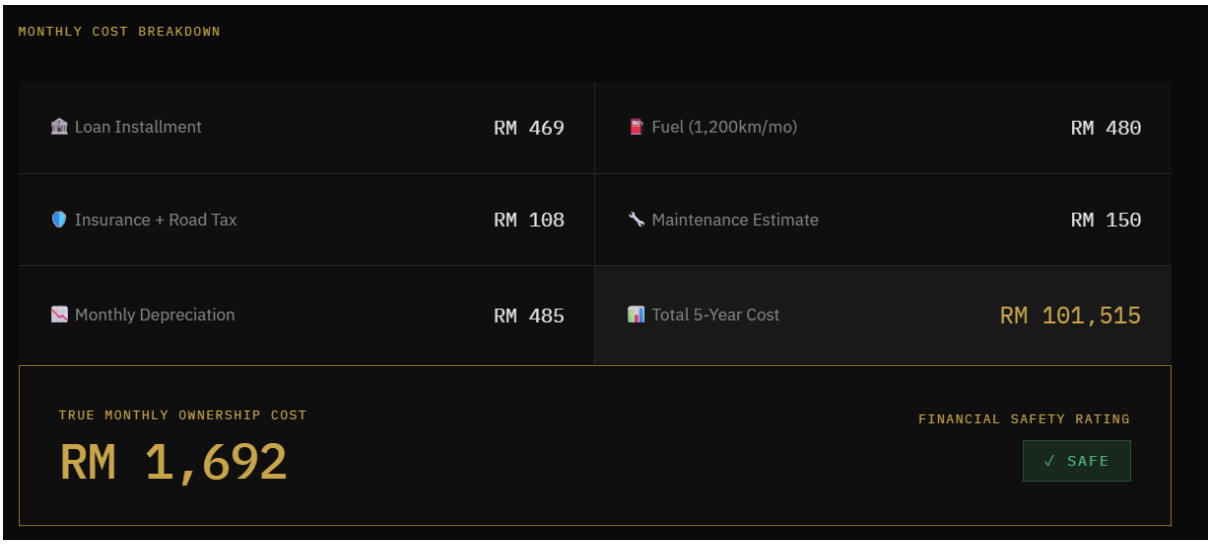
AI Feature	Implementation Details
Car Recommendation Engine (Step 3)	To keep things real and relevant, the app employs a deterministic pre-filter to prevent bogging down the context window. This is done by first extracting cars that are within up to 1.5 times the user's calculated Safe Car Budget. This math-constrained subset of up to 40 cars is then passed along to the LLM along with the user's financial profile. The model then returns a structured JSON array that specifies exactly where the three best matches are, ensuring that math is on the AI's side before any prompting is done.
Full Financial Analysis (Step 6)	Sends the complete financial profile and selected car data. Returns a structured JSON object with fields: verdict, affordability_score, key_risks, safer_alternative, long_term_impact, and executive_summary.
AI Chat (Step 6)	Maintains a rolling 8-message conversation history. The system prompt is dynamically constructed to include the user's salary, DTI, car choice, true monthly cost, and wealth gap figures, enabling contextually grounded advice.

4.0 Results

This section empirically validates TrueCost's viability as a financial intervention tool by evaluating its core objectives, algorithmic efficiency, and AI latency. Through contrasting demographic case studies, it demonstrates the accuracy of the engine's risk-throttling logic and the contextual precision of the AI advisor.

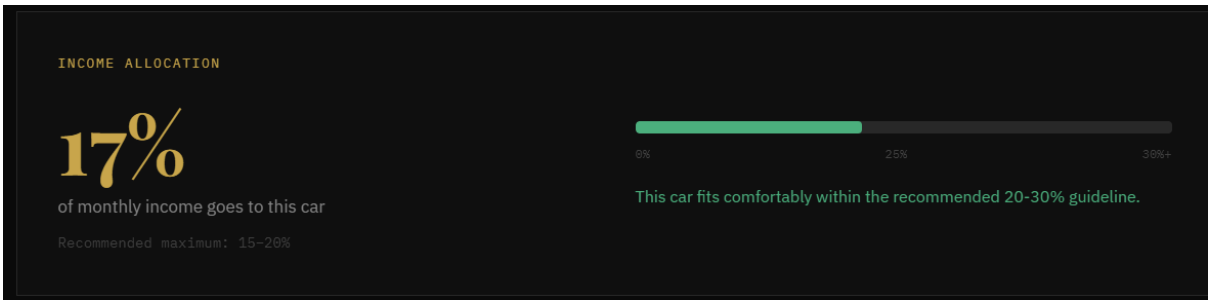
4.1 Objectives Achievement

I. True Monthly Cost Calculation



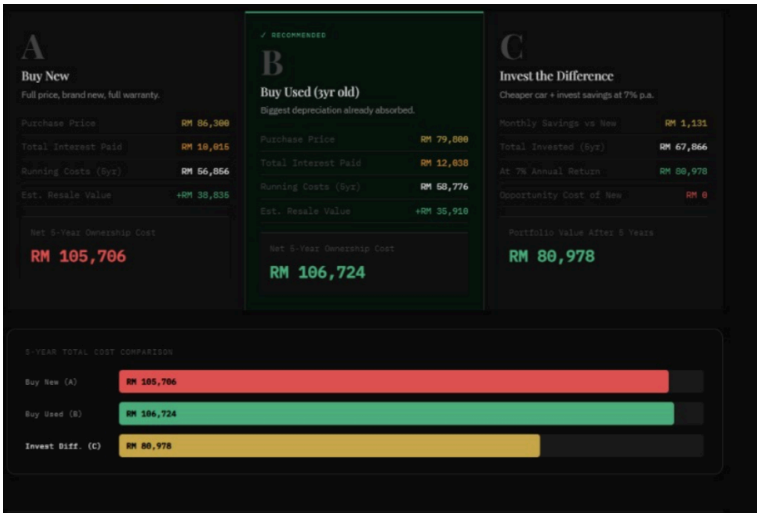
All 5 components (loan, fuel, insurance, maintenance, depreciation) are computed and displayed in Step 4 (True Cost Breakdown). Average revealed cost uplift: ~RM1,200–1,600 above the instalment alone.

II. Personalised Affordability Assessment



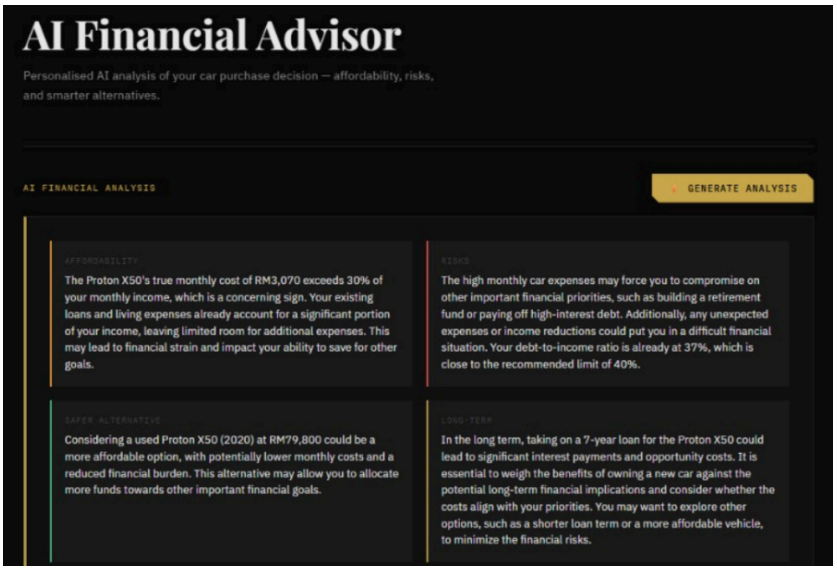
Step 2 (Financial Profile) computes the DTI ratio, safe budget, and maximum affordable car price across 3 loan tenures. Risk badge (SAFE / CAUTION / DANGEROUS) provides immediate visual feedback.

III. Wealth Opportunity Cost Visualisation



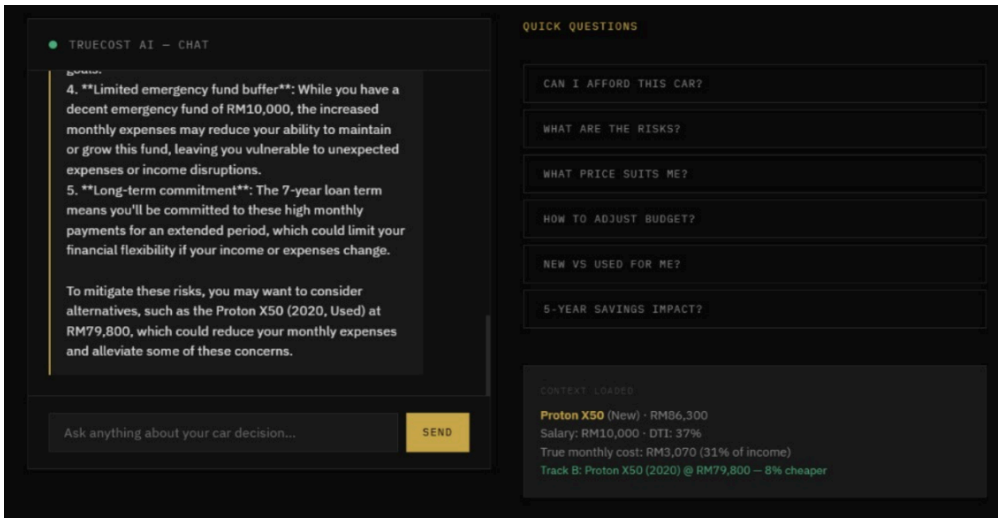
Step 5 (Wealth Gap Analysis) renders interactive SVG charts comparing Track A (chosen car) vs Track B (cheaper alternative + investment) over 5 and 10 years, plus the 3-scenario bar chart.

IV. AI-Powered Financial Advice



Step 6 provides a structured AI analysis and a full context-aware conversational advisor using Groq's llama-3.3-70b-versatile model.

V. Empowerment of Informed Decisions



End-to-end flow from problem awareness (Step 1) to actionable AI advice (Step 6) in a free, zero-registration, single-file web application.

4.2 Technical Performance

The single-page application architecture demonstrated high efficiency during execution:

- **Querying the Dataset:** Applying a filter to the 919-vehicle JSON array on the client side takes virtually no time.
- **AI Inference Latency & Efficiency:** As we filter the dataset ahead of building our prompt, the payload that needs to be sent to Groq API (llama-3. 3-70b-versatile) is kept unconditionally well under common token manifolds. As a result of this

optimization, inference times remain very consistent with between 1.5 - 3 seconds for both vehicle recommendations as well as responses from the interactive chat.

- **State Management:** The good news is that the application maintains state properly across all six screens and passes the calculated variables (DTI, True Cost, Wealth Gap) into context of conversational AI

4.3 System Validation: Contrasting Case Studies To validate the dynamic nature of the TrueCost engine, the system was tested against differing demographic profiles to ensure the logic and AI responded appropriately.

Case Study A: The Standard Profile

- **Inputs**: Salary RM 6,000 | Existing Loans: RM 500 | Dependents: 0 | Job: Stable
- **Selected Car**: Honda Civic 1.5 Turbo (New) — RM 131,900
- **System Output**: The system calculated a Safe Budget of RM 1,200 (unthrottled). The DTI ratio flagged as **CAUTION (38%)**.
- **AI Behaviour**: The AI correctly identified that while the user could technically service the loan, it heavily restricted savings. It successfully recommended Track B alternatives (e.g., a used Honda City at ~RM 70,000) and accurately projected a 10-year wealth gap difference of over RM 84,000 if the monthly savings were invested.

Case Study B: The High-Risk Profile (Testing Demographic Throttling)

- **Inputs**: Salary RM 4,500 | Existing Loans: RM 900 | Dependents: 3 | Job: Freelance
- **Selected Car**: Proton X70 (New) — RM 98,800
- **System Output**: The demographic logic throttled the Safe Budget multiplier down heavily due to freelance status and dependents. The safe budget is computed at just RM 504/month. The DTI ratio flagged as **DANGEROUS (62%)**.
- **AI Behaviour**: The LLM successfully adapted its tone to a stronger warning, explicitly citing the freelance income volatility. The pre-filtering logic restricted the AI from recommending *any* new SUVs, forcing it to suggest reliable, high-depreciation used compact cars (e.g., Perodua Axia at ~RM 30,000) to rescue the user's cash flow.

5.0 Prototype Showcase

5.1 Screen 1 — Landing Page

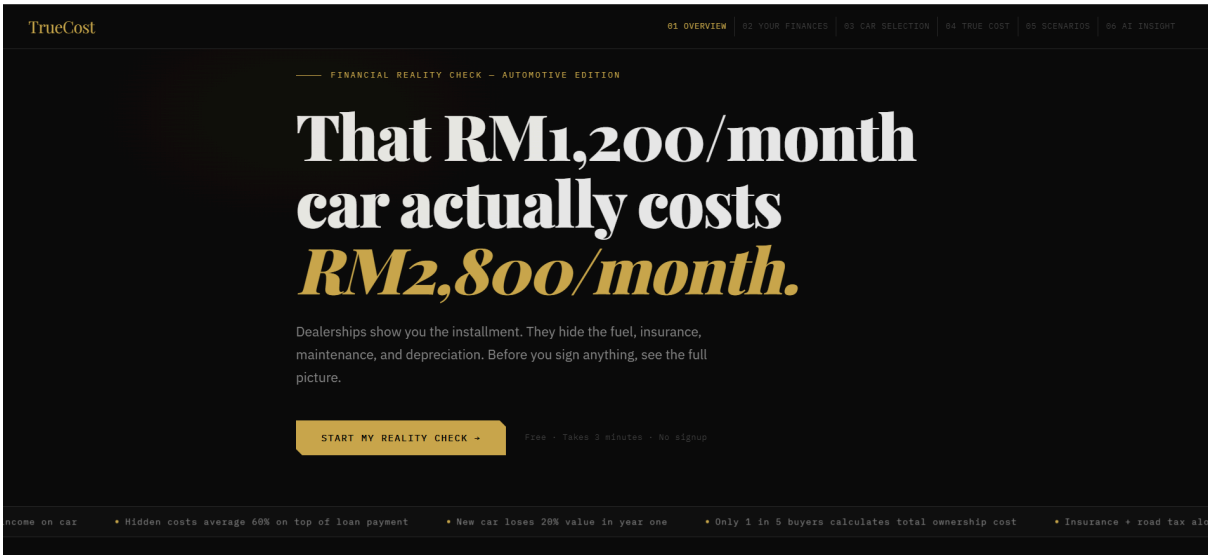


Figure 5.1 creates an immediate emotional impact by revealing that a seemingly affordable RM1,200/month car actually costs RM2,800/month in total. Animated ticker statistics reinforce the problem's scale.

Screen 2 — Financial Profile

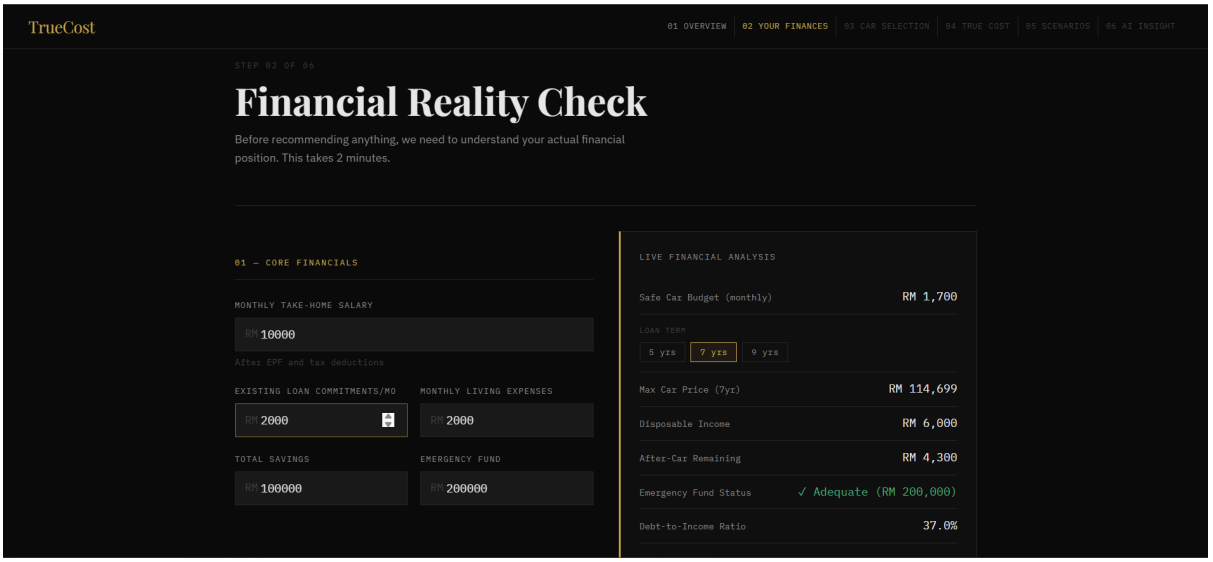


Figure 5.2 collects the user's financial data and immediately computes an affordability verdict, safe budget range, and maximum car prices across three loan tenures (5, 7, and 9 years).

5.3 Screen 3 — Car Selection

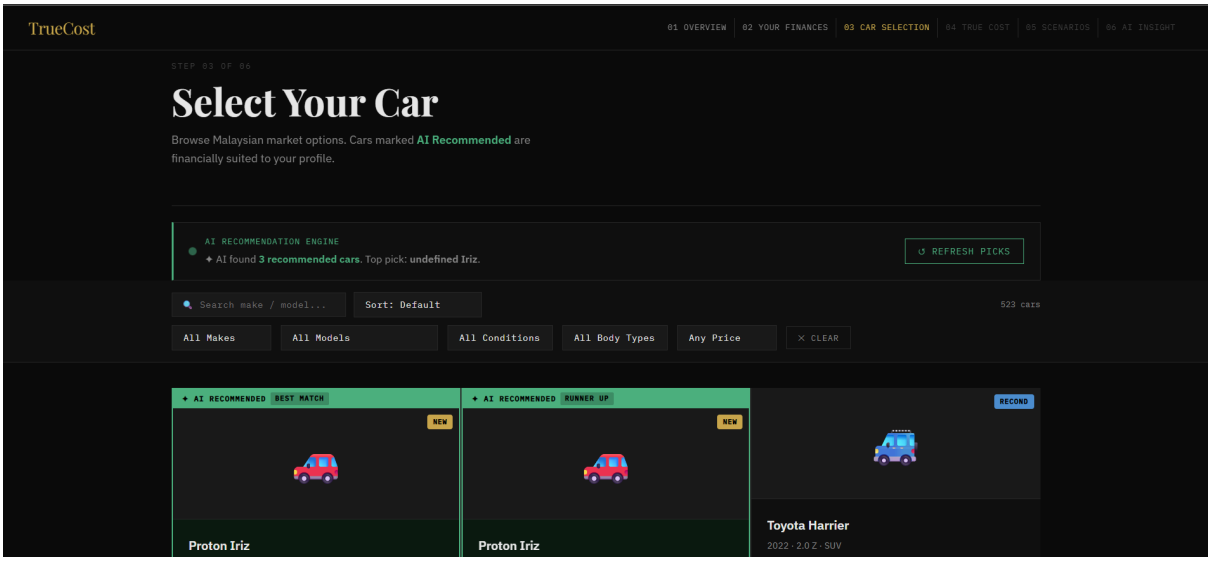


Figure 5.3 presents the 919-vehicle dataset in a browsable, filterable grid with AI-powered recommendations highlighting the three best-matched vehicles for the user's profile.

Screen 4 — True Cost Breakdown

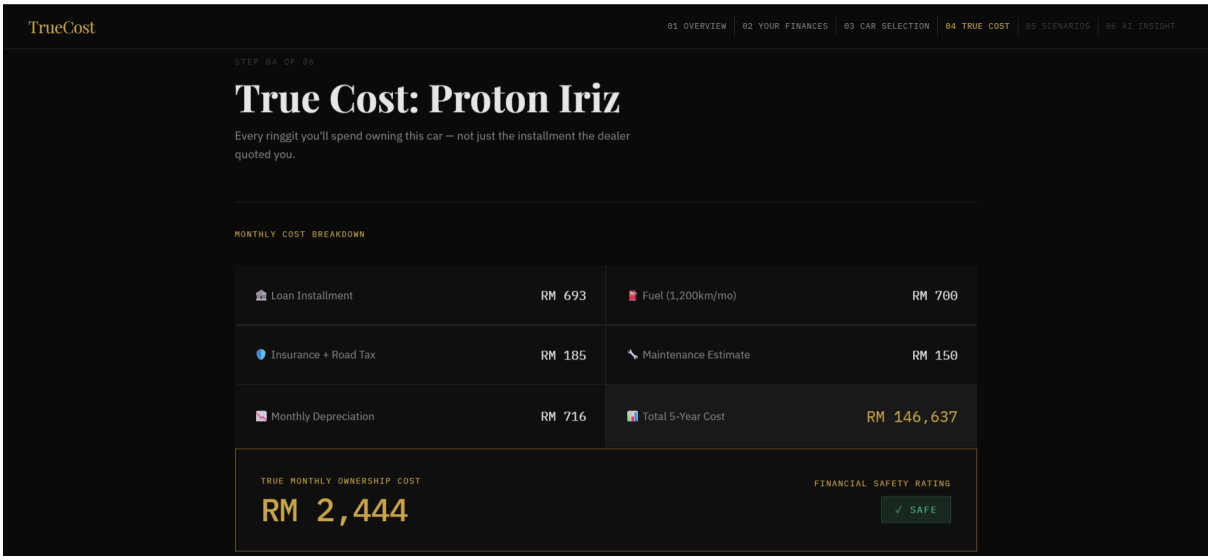


Figure 5.4 presents the full financial reality of the chosen vehicle, breaking the total monthly cost into its five constituent components and projecting the 5-year cumulative spend.

5.5 Screen 5 — Wealth Gap Analysis

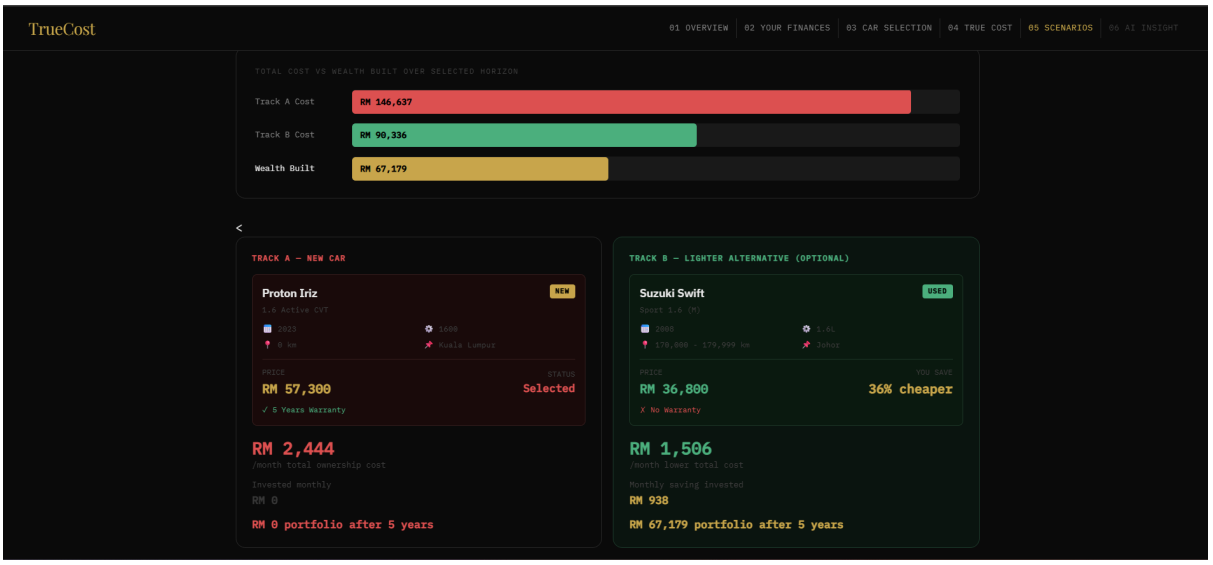


Figure 5.5 quantifies the long-term wealth impact of the car choice, showing users what they could have accumulated by choosing a less expensive vehicle and investing the difference.

5.6 Screen 6 — AI Financial Advisor

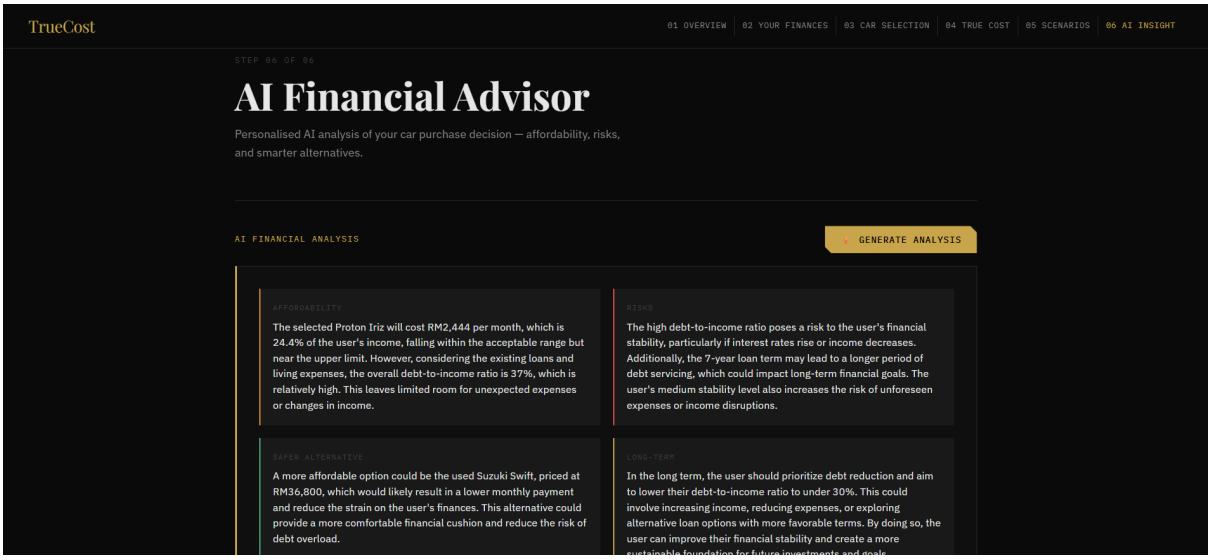


Figure 5.6 delivers a comprehensive AI-generated financial assessment of the user's car choice, followed by a context-aware conversational advisor for follow-up questions.

6.0 AI Acknowledgment

TrueCost uses Claude to assist with development, debugging, and formula implementation, while Groq API powers vehicle recommendations and AI financial advice, with all outputs reviewed and validated by humans for accuracy and reliability.

6.1 AI Tools Used in Development

AI Tool	Role in This Project
Claude (Antrophic)	Assisting in the calculation formula implementation and debugging when errors occur
Groq API (llama-3.3-70b-versatile)	Integrated into the application as the AI Inference Engine powering two features: 1. Car Recommendation Engine (Step 3) 2. AI Financial Advisor (Step 6)

6.2 Specific use of Claude (Antropic)

Claude was used as a development assistant during the creation of the TrueCost. It supports rapid prototyping, debugging, and refinement throughout the development process. Specific uses included:

- Supporting the implementation and validation of financial calculation formulas (DTI ratio, true cost components, and wealth gap compound interest model) in JavaScript.
- Helping construct dynamic system prompt templates that inject user financial context into Groq API requests.
- Assisting with the development of SVG-based chart rendering functions used for the Track A vs Track B wealth comparison visualisation.

6.3 Specific Use of Groq API

The Groq API serves as an integral runtime component of the deployed application rather than merely a development tool. It is utilised in three primary operational contexts:

- **Car Recommendation Engine:** At runtime, the top 40 vehicles filtered according to the user's selection criteria are submitted to the *llama-3.3-70b-versatile* model together with the user's financial profile. The model returns a structured JSON array identifying the three best-matched vehicle IDs, accompanied by plain-language justifications for each recommendation.
- **AI Financial Advisor:** The user's complete financial profile, selected vehicle details, computed true monthly cost, and projected wealth gap metrics are provided to the model. In response, the model generates a structured JSON object containing an

affordability verdict, key financial risk factors, a safer alternative suggestion, a long-term financial outlook, and a concise executive summary.

6.4 Human Oversight

Human oversight measures applied during development:

- All AI-generated code was reviewed, tested, and validated by human developers before inclusion.
- Financial Formulas were cross-referenced against Bank Negara Malaysia (BNM) guidelines and publicly available actuarial data.
- The vehicle dataset was verified against the current Malaysian market pricing from authorized sources.
- AI-generated financial advice output is clearly labelled as AI-generated within the application interface.

7.0 Conclusion

7.1 Project Summary

TrueCost — *Car Affordability Reality Check* was developed to address the need for accessible financial decision-support tools in an increasingly complex digital financial ecosystem. As digital financial services expand across ASEAN, many consumers still struggle to understand the long-term financial impact of major purchases such as vehicles.

The application provides a browser-based platform that integrates personal financial profiling, true-cost modelling, wealth-opportunity-cost analysis, and AI-powered financial advisory services to help users evaluate vehicle affordability more realistically.

TrueCost promotes financial awareness and responsible decision-making, particularly for individuals in the B40 and M40 income groups, by simplifying complex financial calculations and presenting them in an intuitive format.

Aligned with SDG 1, SDG 8, and SDG 10, the system aims to strengthen financial literacy and empower consumers to make more informed long-term financial decisions.

7.2 Future Improvements

Future development will focus on improving the accuracy, accessibility, and user experience:

i. Live Data Integration Real-time vehicle pricing from platforms such as Carlist.my or Mudah.my.
ii. Dataset Expansion – Increase coverage beyond the current 919 vehicles.
iii. User Profiles & History – Enable optional accounts to save financial profiles and comparisons.
iv. EPF Planning Module – Model the long-term retirement impact of using EPF Account 2 for vehicle down payments.
v. Vehicle Expenditure Tracker – Track actual monthly spending against projected true cost.

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